```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
```

```
import kagglehub

# Download latest version
path = kagglehub.dataset_download("krishnaraj30/mall-visiting-customer-data")

print("Path to dataset files:", path)
```

```
Downloading to /root/.cache/kagglehub/datasets/krishnaraj30/mall-visiting-customer-data/1.archive...
100%|██████████| 1.55k/1.55k [00:00<00:00, 1.93MB/s]Extracting files...
Path to dataset files: /root/.cache/kagglehub/datasets/krishnaraj30/mall-visiting-customer-data/versions/1
```

```
import os
print(os.listdir(path))
df = pd.read_csv(os.path.join(path, "Mall_Customers.csv"))
df.head()
```

```
['Mall_Customers.csv']
```

	CustomerID	Gender	Age	Annual Income (k$)	Spending Score (1-100)
0	1	Male	19	15	39
1	2	Male	21	15	81
2	3	Female	20	16	6
3	4	Female	23	16	77
4	5	Female	31	17	40

1.Basic EDA

```
print(df.shape)
```

```
(200, 5)
```

```
df.columns
```

```
Index(['CustomerID', 'Gender', 'Age', 'Annual Income (k$)',
       'Spending Score (1-100)'],
      dtype='object')
```

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 200 entries, 0 to 199
Data columns (total 5 columns):
 #   Column                  Non-Null Count  Dtype
---  ------                  --------------  -----
 0   CustomerID              200 non-null    int64
 1   Gender                  200 non-null    object
 2   Age                     200 non-null    int64
 3   Annual Income (k$)      200 non-null    int64
 4   Spending Score (1-100)  200 non-null    int64
dtypes: int64(4), object(1)
memory usage: 7.9+ KB
```

```
df.describe()
```

	CustomerID	Age	Annual Income (k$)	Spending Score (1-100)
count	200.000000	200.000000	200.000000	200.000000
mean	100.500000	38.850000	60.560000	50.200000
std	57.879185	13.969007	26.264721	25.823522
min	1.000000	18.000000	15.000000	1.000000
25%	50.750000	28.750000	41.500000	34.750000
50%	100.500000	36.000000	61.500000	50.000000
75%	150.250000	49.000000	78.000000	73.000000
max	200.000000	70.000000	137.000000	99.000000

```
df.isnull().sum()
```

	0
CustomerID	0
Gender	0
Age	0
Annual Income (k$)	0
Spending Score (1-100)	0

dtype: int64

```
df.duplicated().sum()
```

```
np.int64(0)
```

```
df['Age'] = df['Age'].fillna(df['Age'].mean())
```

```
df['Annual Income (k$)'] = df['Annual Income (k$)'].fillna(df['Annual Income (k$)'].median())
```

Data cleaning

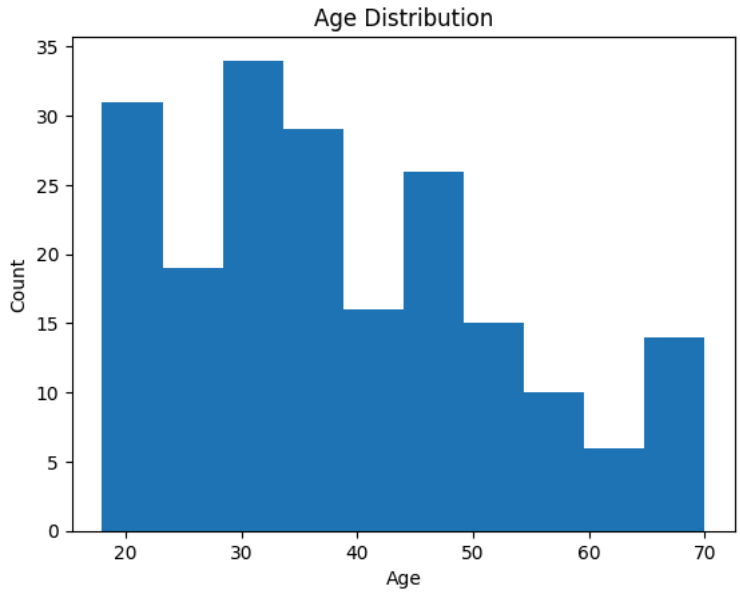
```
# Convert Gender to numeric
from sklearn.preprocessing import LabelEncoder
le = LabelEncoder()
df['Gender'] = le.fit_transform(df['Gender'])
print("DataFrame after Gender conversion:")
display(df.head())
```

```
DataFrame after Gender conversion:
```

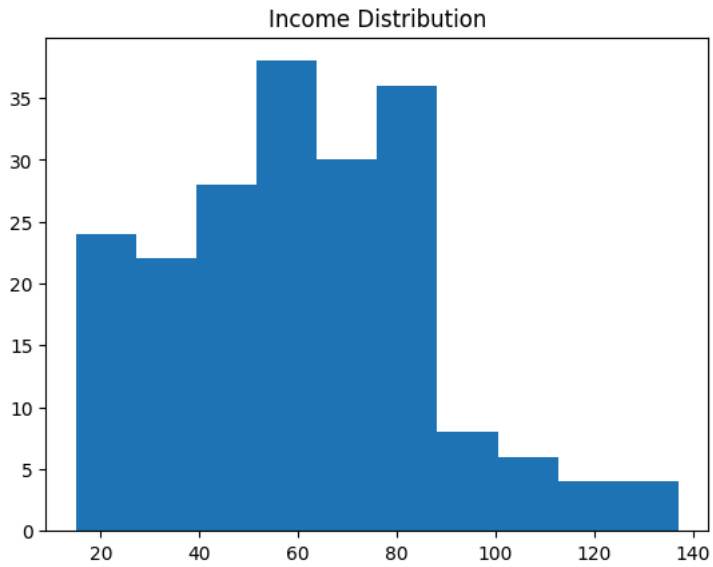
	CustomerID	Gender	Age	Annual Income (k$)	Spending Score (1-100)
0	1	1	19	15	39
1	2	1	21	15	81
2	3	0	20	16	6
3	4	0	23	16	77
4	5	0	31	17	40

Univariate Analysis (Single Column)

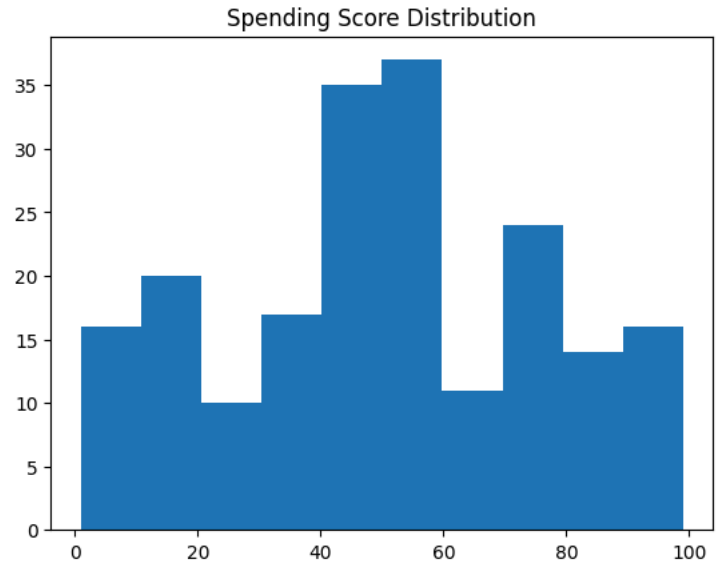
```
#Age Distribution
plt.hist(df['Age'])
plt.title("Age Distribution")
plt.xlabel("Age")
plt.ylabel("Count")
plt.show()
```




```
# Income Distribution
plt.hist(df['Annual Income (k$)'])
plt.title("Income Distribution")
plt.show()
```




```
# Spending Score
plt.hist(df['Spending Score (1-100)'])
plt.title("Spending Score Distribution")
plt.show()
```

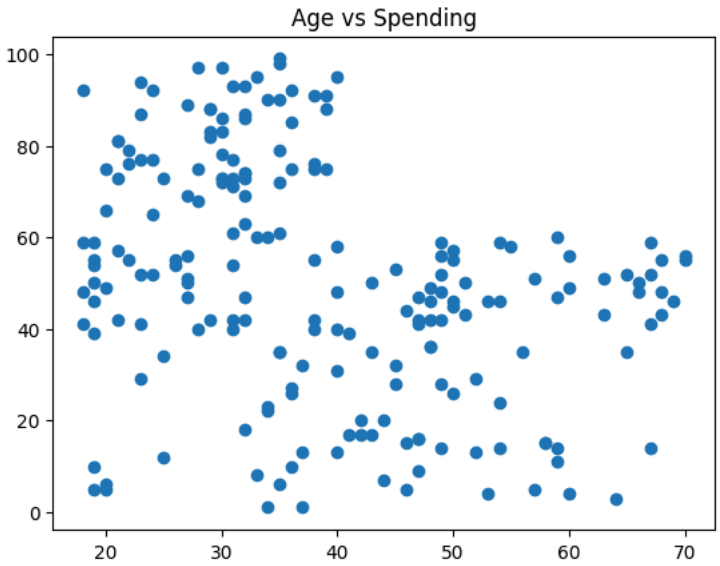


Start coding or generate with AI.

Bivariate Analysis (Relationship)

```
#Income vs Spending Score
plt.scatter(df['Annual Income (k$)'], df['Spending Score (1-100)'])
plt.xlabel("Income")
plt.ylabel("Spending Score")
plt.title("Income vs Spending")
plt.show()
```

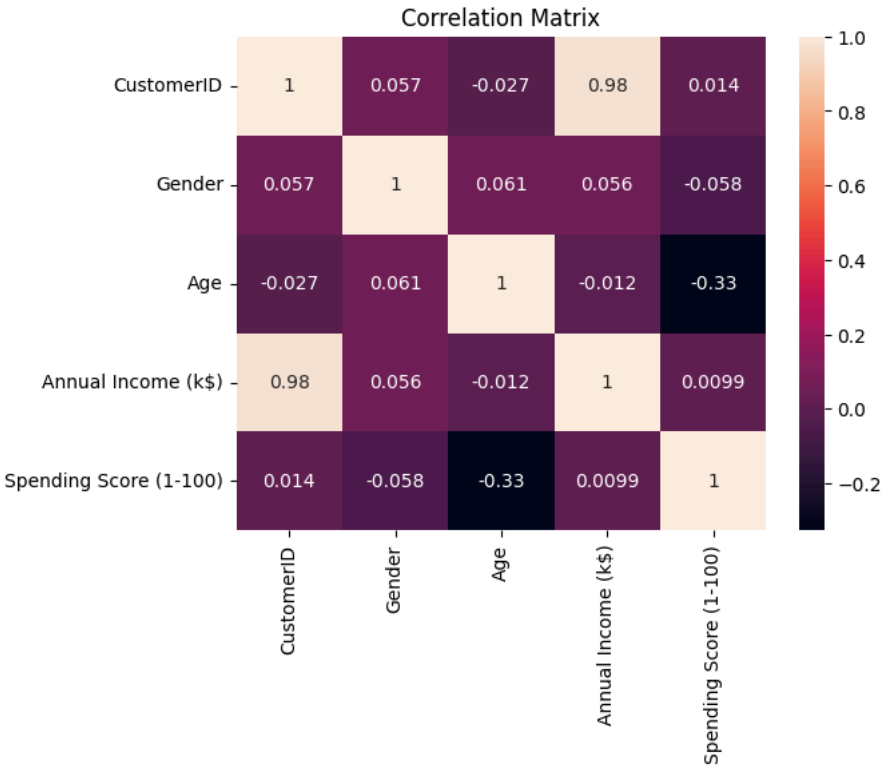
```
# Age vs Spending
plt.scatter(df['Age'], df['Spending Score (1-100)'])
plt.title("Age vs Spending")
plt.show()
```



Start coding or generate with AI.

Correlation Matrix

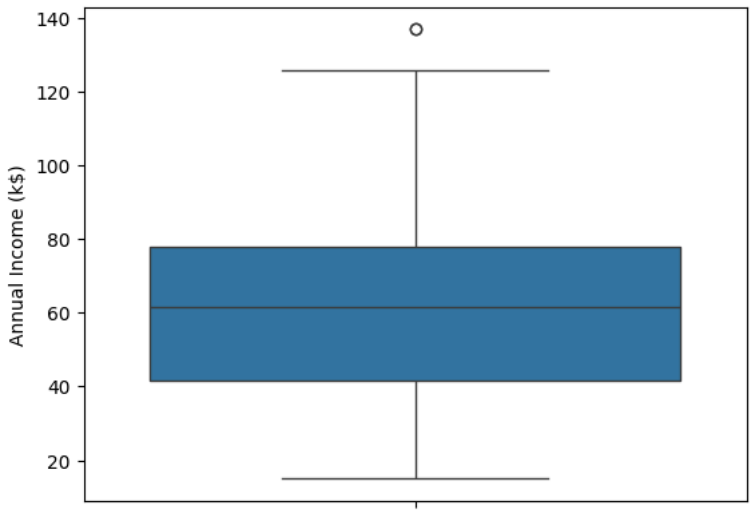
```
sns.heatmap(df.corr(), annot=True)
plt.title("Correlation Matrix")
plt.show()
```

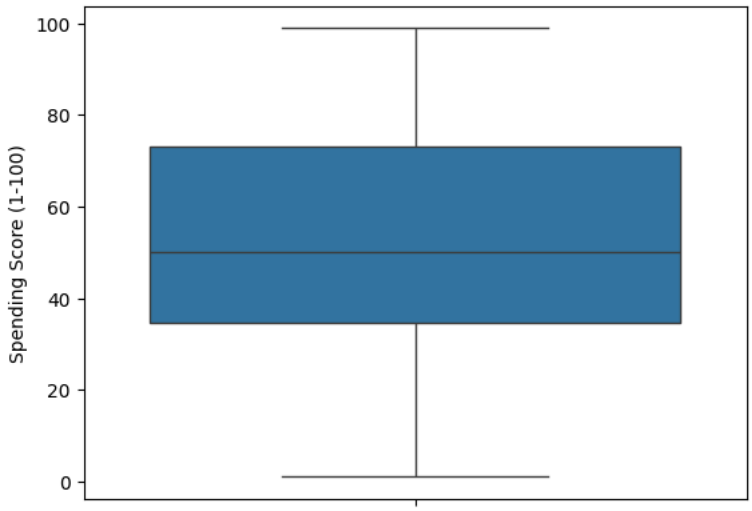


Outlier Detection (Boxplot)

```
sns.boxplot(df['Annual Income (k$)'])
plt.show()

sns.boxplot(df['Spending Score (1-100)'])
plt.show()
```





Start coding or generate with AI.

Model Building

Feature selection

```
X = df[['Age', 'Annual Income (k$)', 'Spending Score (1-100)']]
```

Feature Scaling

```
from sklearn.preprocessing import StandardScaler

scaler = StandardScaler()
X_scaled = scaler.fit_transform(X)
```

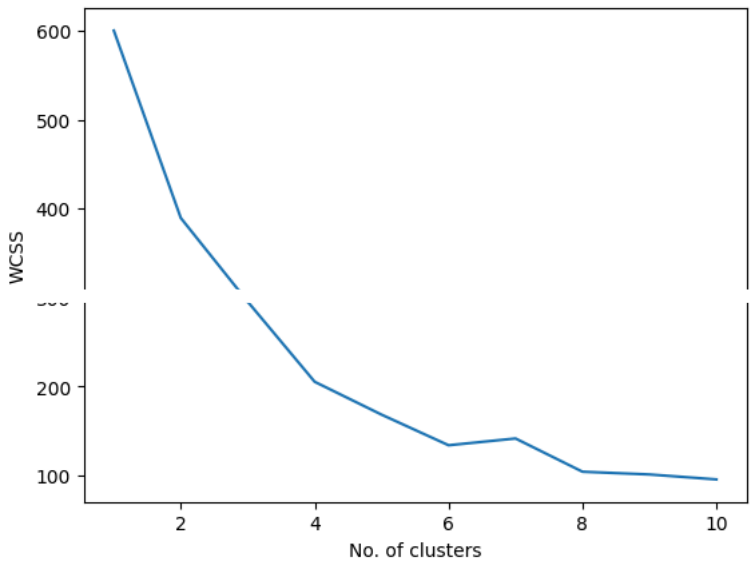
Find Optimal K (Elbow Method)

```
from sklearn.cluster import KMeans
import matplotlib.pyplot as plt

wcss = []
for i in range(1, 11):
    kmeans = KMeans(n_clusters=i)
    kmeans.fit(X_scaled)
    wcss.append(kmeans.inertia_)

plt.plot(range(1,11), wcss)
plt.xlabel("No. of clusters")
```

```
plt.ylabel("WCSS")
```

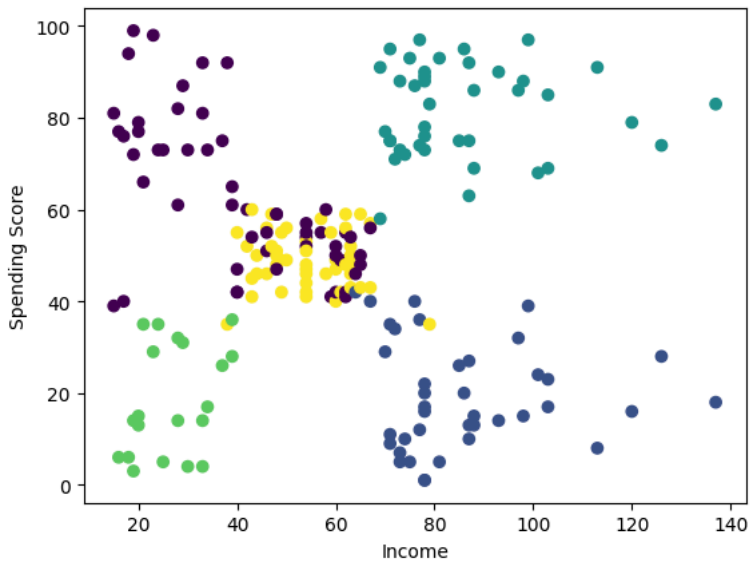


Apply K-Means Model

```
kmeans = KMeans(n_clusters=5)
df['Cluster'] = kmeans.fit_predict(X_scaled)
```

Visualization

```
plt.scatter(df['Annual Income (k$)'], df['Spending Score (1-100)'], c=df['Cluster'])
plt.xlabel("Income")
plt.ylabel("Spending Score")
plt.show()
```



Interpretation